

A Technical Introduction to Receiver operating characteristic

Section 1

The Total Operating Characteristic (TOC) also characterizes diagnostic ability while revealing more information than the ROC. For each threshold, ROC reveals two ratios, $TP/(TP + FN)$ and $FP/(FP + TN)$. In other words, ROC reveals hits hits + misses $\{\frac{\text{hits}}{\text{hits} + \text{misses}}\}$ and false alarms false alarms + correct rejections $\{\frac{\text{false alarms}}{\text{false alarms} + \text{correct rejections}}\}$. On the other hand, TOC shows the total information in the contingency table for each threshold. The TOC method reveals all of the information that the ROC method provides, plus additional important information that ROC does not reveal, i.e. the size of every entry in the contingency table for each threshold. TOC also provides the popular AUC of the ROC.

Section 2

$P_D = \frac{1}{2} \operatorname{erfc} \left(\frac{\operatorname{erfc}^{-1}(2P_{FA}) - X}{\sqrt{2}} \right)$

Section 3

G_1 is referred to as Gini index or Gini coefficient, but it should not be confused with the measure of statistical dispersion that is also called Gini coefficient. G_1 is a special case of Somers' D. It is also common to calculate the Area Under the ROC Convex Hull (ROC AUCC = ROCH AUC) as any point on the line segment between two prediction results can be achieved by randomly using one or the other system with probabilities proportional to the relative length of the opposite component of the segment. It is also possible to invert concavities – just as in the figure the worse solution can be reflected to become a better solution; concavities can be reflected in any line segment, but this more extreme form of fusion is much more likely to overfit the data. The machine learning community most often uses the ROC AUC statistic for model comparison. This practice has been questioned because AUC estimates are quite noisy and suffer from other problems. Nonetheless, the coherence of AUC as a measure of aggregated classification performance has been vindicated, in terms of a uniform rate distribution, and AUC has been linked to a number of other performance metrics such as the Brier score. Another problem with ROC AUC is that reducing the ROC Curve to a single number ignores the fact that it is about the tradeoffs between the different systems or performance points plotted and not the performance of an individual system, as well as ignoring the possibility of concavity

repair, so that related alternative measures such as Informedness or DeltaP are recommended. These measures are essentially equivalent to the Gini for a single prediction point with $\Delta P' = \text{Informedness} = 2\text{AUC} - 1$, whilst $\Delta P = \text{Markedness}$ represents the dual (viz. predicting the prediction from the real class) and their geometric mean is the Matthews correlation coefficient. Whereas ROC AUC varies between 0 and 1 — with an uninformative classifier yielding 0.5 — the alternative measures known as Informedness, Certainty and Gini Coefficient (in the single parameterization or single system case) all have the advantage that 0 represents chance performance whilst 1 represents perfect performance, and -1 represents the "perverse" case of full informedness always giving the wrong response. Bringing chance performance to 0 allows these alternative scales to be interpreted as Kappa statistics. Informedness has been shown to have desirable characteristics for Machine Learning versus other common definitions of Kappa such as Cohen Kappa and Fleiss Kappa. Sometimes it can be more useful to look at a specific region of the ROC Curve rather than at the whole curve. It is possible to compute partial AUC. For example, one could focus on the region of the curve with low false positive rate, which is often of prime interest for population screening tests. Another common approach for classification problems in which $P \ll N$ (common in bioinformatics applications) is to use a logarithmic scale for the x-axis. The ROC area under the curve is also called c-statistic or c statistic.

Section 4

A receiver operating characteristic curve, or ROC curve, is a graphical plot that illustrates the performance of a binary classifier model (although it can be generalized to multiple classes) at varying threshold values. ROC analysis is commonly applied in the assessment of diagnostic test performance in clinical epidemiology. The ROC curve is the plot of the true positive rate (TPR) against the false positive rate (FPR) at each threshold setting. The ROC can also be thought of as a plot of the statistical power as a function of the Type I Error of the decision rule (when the performance is calculated from just a sample of the population, it can be thought of as estimators of these quantities). The ROC curve is thus the sensitivity as a function of false positive rate. Given that the probability distributions for both true positive and false positive are known, the ROC curve is obtained as the cumulative distribution function (CDF, area under the probability distribution from $-\infty$ to the discrimination threshold) of the detection probability in the y-axis versus the CDF of the false positive probability on the x-axis. ROC analysis provides tools to select possibly optimal models and to discard suboptimal ones independently from (and prior to specifying) the cost context or the class distribution. ROC analysis is related in a direct and natural way to the cost/benefit analysis of diagnostic decision making.

Section 5

Terminology The true-positive rate is also known as sensitivity or probability of detection. The false-positive rate is also known as the probability of false alarm and equals $(1 - \text{specificity})$. The ROC is also known as a relative operating characteristic curve, because it is a comparison of two operating characteristics (TPR and FPR) as the criterion changes.

Section 6

the intercept of the ROC curve with the line at 45 degrees orthogonal to the no-discrimination line - the balance point where Sensitivity = Specificity the intercept of the ROC curve with the tangent at 45 degrees parallel to the no-discrimination line that is closest to the error-free point (0,1) – also called Youden's J statistic and generalized as Informedness the area between the ROC curve and the no-discrimination line multiplied by two is called the Gini coefficient, especially in the context of credit scoring. It should not be confused with the measure of statistical dispersion also called Gini coefficient. the area between the full ROC curve and the triangular ROC curve including only (0,0), (1,1) and one selected operating point (tpr, fpr) – Consistency the area under the ROC curve, or "AUC" ("area under curve"), or A' (pronounced "a-prime"), or "c-statistic" ("concordance statistic"). the sensitivity index d' (pronounced "d-prime"), the distance between the mean of the distribution of activity in the system under noise-alone conditions and its distribution under signal-alone conditions, divided by their standard deviation, under the assumption that both these distributions are normal with the same standard deviation. Under these assumptions, the shape of the ROC is entirely determined by d' . However, any attempt to summarize the ROC curve into a single number loses information about the pattern of tradeoffs of the particular discriminator algorithm.